

Answer the following questions below. This is based on what you have learned already.

- 1: The equation for calculating density is?
- 2: Density is defined as the mass per unit
3. Which state of matter has the highest density?
- 4: What is density of an object with a mass of 10 kg and volume of 2 m³ ?
5. What is the mass of an object if it has a density of 2 g/cm³ and a volume of 60 cm³?
- 6: If an object has a density of 1200 kg/m³ what is it's density in g/cm³

- 1: The equation for calculating density is? **Density = mass / volume**
- 2: Density is defined as the mass per unit **Volume**
3. Which state of matter has the highest density? **Solid**
- 4: What is density of an object with a mass of 10 kg and volume of 2 m³ ? **5 kg/m³**
- 5: What is the mass of an object if it has a density of 2 g/cm³ and a volume of 60 cm³? **120 g**
- 6: If an object has a density of 1200 kg/m³ what is it's density in g/cm³ **1.2**

Quiz on BBC bitesize

3.2 Required practical

To investigate the density of regular and irregular solid objects and liquids



Queen
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Density

Density is the amount of mass in a specified space. It is a way to measure how tightly packed particles are in an object.

Why do icebergs float?

The simple answer is: ice is **less dense** than water. When water freezes, it expands - this means its mass remains the same but its volume has changed, altering its density.



You will be carrying out three experiments that will enable you to find out the density of different objects.

You must know the following for us to continue

density of a material is found using the equation

$$\text{Density} = \frac{\text{mass}}{\text{Volume}}$$

$$\rho = \frac{m}{V}$$

Density ρ in kilograms per metre cubed kg/m^3

Volume V in metres cubed m^3

The different ways to measure the density of regular and irregular solids

Who was it that discovered the way to measure the volume of an irregular shaped solid?

There are different ways to investigate density. In this required practical activity, however, it is important to:

- record the mass accurately
- measure and observe the mass and the volume of the different objects
- use appropriate apparatus and methods to measure volume and mass

The Experiments

Investigating the density of regular objects,
irregular objects and liquids.



Experiment 1

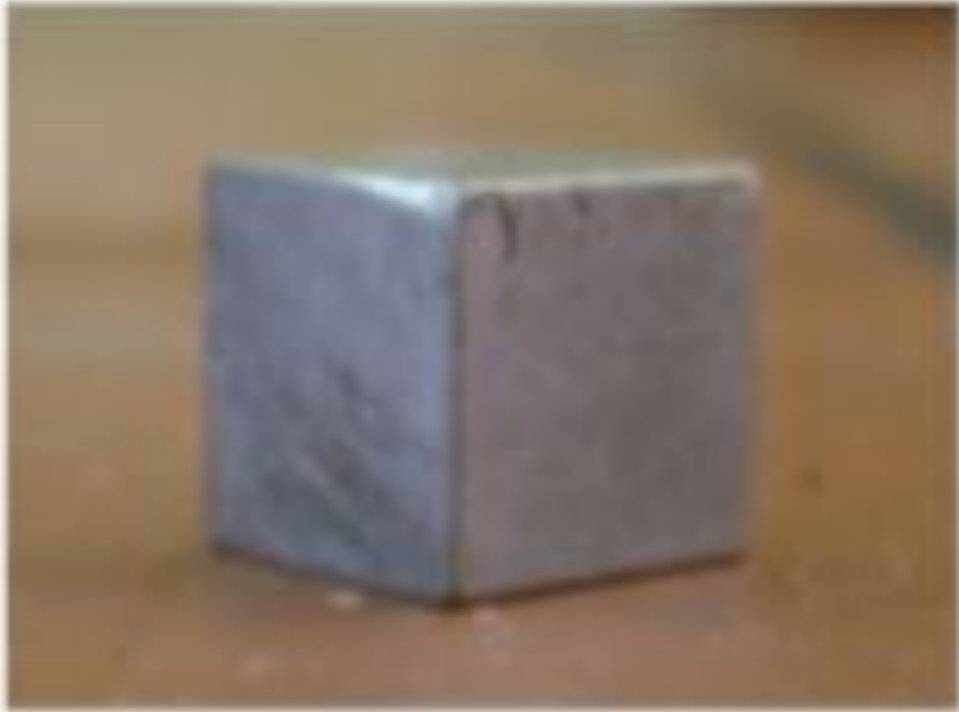
The Density of Regular Objects

Aim

To identify the material an object is made from by calculating its density.



How would you determine the mass and volume of the metal cube below?



Density equation

Density = mass/volume

$$\rho = \frac{m}{V}$$

Units:

Mass - kg

Volume - m³

Density - kg/m³

Mass - g

Volume - cm³

Density - g/cm³

Finding the density of regular solid objects

Equipment	What it is used for
Metal cube	To determine its density
Ruler	To measure the length of each side of the cube in cm
Top pan balance	To measure the mass of the cube in g

Hazards and control measures

Hazard	Consequence	Control measures
Water spilled from displacement can	Slip and fall	Use a measuring cylinder to collect displaced water and prevent spills

Aim of the experiment

To measure the density of various materials. Regular solid objects

Method

1. Use a ruler to measure the length (l), width (w) and height (h) of a cube.
2. Place the cube on the top pan balance and measure its mass.
3. Calculate the volume of the cube using $(l \times w \times h)$.
4. Use the measurements to calculate the density of the metal.

You must know the following for us to continue

Available equipment

- Safety goggles(must be worn at all times)
- Mass balance
 - (Weighing scales)
- Eureka can
- Ruler
- Measuring cylinder
- Water

Method 2: Stone or other Irregular shaped solid objects

Method

1. Place the stone on the top pan balance and measure its mass.
2. Fill the displacement can until the water is level with the bottom of the pipe.
3. Place a measuring cylinder under the pipe ready to collect the displaced water.
4. Carefully drop the stone into the can and wait until no more water runs into the cylinder.
5. Measure the volume of the displaced water.
6. Use the measurements to calculate the density of the stone.

Method 3: Water (or any liquid)

1. Place the measuring cylinder on the top pan balance and measure its mass.
2. Pour 50 cm^3 of water into the measuring cylinder and measure its new mass.
3. Subtract the mass in step 1 from the mass in step 2. This is the mass of 50 cm^3 of water.
4. Use the measurements to calculate the density of the water.

Results table

Finding the density of irregular solid objects

Equipment	What it is used for
Rock	To determine its density
Eureka Can	To measure the volume of the irregular object
Measuring Cylinder	To measure the volume of the irregular object in cm^3
Top pan balance	To measure the mass of the irregular object in g

Exam question

Below is a picture of an irregular solid object. A student wants to find the density of the object.



Image credit: wikimedia, Evan-Amos, Standard lock key

Plan an experiment that would allow the student to determine the density of the object. **(6)**

Conclusion

- What do your results show you?

Evaluation

- What was the biggest error with your experiment?
-
- How would you try to overcome this error?
- What would you keep the same?
- Are your results reliable?